Optimization Models for Targeted Offers in Direct Marketing

Implementation and computational study of the exact formulations

after F. Talla Nobibon, R. Leus, F.C.R. Spieksma, *EJOR* 210 (2011) 670–683

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Homework 3

Problem and models follow [1]. **Blue** *marks the presenter's implementation, experiments and interpretation.*

Outline

1. Problem and data
2. Formulations
3. Implementation
4. Experimental design
5. Experiment 1: small instances
6. Experiment 2: groups S1 to S3
7. Summary

The problem

- A **direct-marketing campaign** offers products to selected customers. The paper addresses *retention* campaigns, which seek additional revenue from existing customers of a bank [1].
- Response models estimate the expected return $p_{ij} = r_{ij}\, DFV_{ij}$ of offering product j to customer i (acceptance probability times value to the firm); these estimates are **inputs** of the model.

Product targeting problem

Decide which products to offer to which customers so as to maximize the expected net profit of the campaign, subject to four business requirements:

1. a corporate **hurdle rate** R on the return on investment, ;
2. a **budget** B_j per product, because each business unit funds its own product;
3. a maximum number M_i of **offers per customer**;
4. a **minimum-quantity commitment** (MQC) O_j: a product used in the campaign is offered to at least O_j customers.

A **fixed cost** f_j is incurred whenever product j is used.

The problem is strongly NP-hard; even finding a solution with positive profit is NP-hard [1].

Implementation: Instance generation

Generation rules of Section 5.1 of [1], compared with the implementation. Notation: $C_j = \sum_i c_{ij}$, $\bar{M} = \sum_i M_i/n$, $F_j = \frac{O_j}{m(1+R)} \sum_i \left[p_{ij} - (1+R)c_{ij}\right]$.

Parameter	Meaning	Rule in the paper	Implementation	Check
c_{ij}	cost of an offer	integer, uniform in $\{1, 2, 3\}$	same	✓
p_{ij}	return of an offer	integer, uniform in $\{0, \ldots, 16\}$	same	✓
R	hurdle rate	5%, 10% or 15%	same	✓
m, n	customers, products	$m = 100, 200, 300$; $n = 5, 10, 15$	same	✓
M_i (s)	offers per customer	uniform in $\{1, \ldots, n/5\}$	uniform in $\{1, \ldots, \max(1, \lfloor n/5 \rfloor)\}$	✓
M_i (ℓ)	offers per customer	uniform in $\{\lceil n/3 \rceil, \ldots, \lceil 2n/3 \rceil\}$	same	✓
O_j	MQC	uniform in $\{\lceil \bar{M} \rceil, \ldots, \lceil 2\bar{M} \rceil\}$	uniform in $\{\lceil \bar{M} \rceil, \ldots, \lfloor 2\bar{M} \rfloor\}$	**deviation**
B_j (min, max)	budget	$\lfloor O_j C_j/m \rfloor$ and $\lceil 2\bar{M} C_j/m \rceil$	same	✓
B_j (random)	budget	integer between $O_j C_j/m$ and $2C_j/n$	integer between the min and max budgets	**deviation**
f_j	fixed cost	integer between $F_j/2$ and F_j (rounding not stated)	$\lceil F_j/2 \rceil$ to $\lfloor F_j \rfloor$; fallback if empty	✓

Validation of the generator

- Both deviations are documented (O_j cap avoids a minimum budget above the maximum; $2C_j/n$ read as a typo). All 212 instance files are reproduced exactly and satisfy these rules; the f_j fallback is never used.
- **Undocumented:** the random stream restarts with the same seed in every call. For a given n, instances with different R and M_i variant share p_{ij}, c_{ij} (same M_i variant: also M_i, O_j); the first 100 (200) customers of S2 (S3) have the costs of S1 (S2). The paper does not say whether its instances are independent.

Model M1

Variables: $x_{ij} = 1$ if product j is offered to customer i; $y_j = 1$ if product j is used in the campaign.

$$\max \sum_{i,j} (p_{ij} - c_{ij})\, x_{ij} - \sum_j f_j\, y_j \qquad (1)\ \text{net profit}$$

$$\text{s.t.} \sum_{i,j} p_{ij} x_{ij} \;\geq\; (1+R)\Big[\sum_{i,j} c_{ij} x_{ij} + \sum_j f_j y_j\Big] \qquad (2)\ \text{hurdle rate}$$

$$\sum_i c_{ij} x_{ij} \leq B_j, \qquad j = 1, \ldots, n \quad (3)\ \text{budget}$$

$$\sum_j x_{ij} \leq M_i, \qquad i = 1, \ldots, m \quad (4)\ \text{offers per customer}$$

$$\sum_i x_{ij} \leq m\, y_j, \qquad j = 1, \ldots, n \quad (5)\ \text{linking}$$

$$\sum_i x_{ij} \geq O_j\, y_j, \qquad j = 1, \ldots, n \quad (6)\ \text{MQC}$$

$$x_{ij},\ y_j \in \{0, 1\}. \qquad (7)$$

- (M1) has $mn + n$ binary variables and $1 + m + 3n$ constraints. Constraints (5) and (6) together state that a product is either unused or offered to at least O_j customers.
- In the LP relaxation, (5) $\sum_i x_{ij} \leq m\, y_j$ allows $y_j = \sum_i x_{ij}/m$, so the fixed cost f_j is charged only at a rate of only f_j/m per offer. This weakens the LP bound of (M1).

Model M2

(M2) replaces the aggregated linking constraints (5) $\sum_i x_{ij} \leq m\, y_j$ by their **disaggregated** form

$$x_{ij} \leq y_j, \qquad i = 1, \ldots, m,\ j = 1, \ldots, n. \qquad (8)$$

Proposition 3 [1]

The convex hull of the feasible solutions of (M1) equals the convex hull of the solutions satisfying (2)–(4), (6) and (8), with $0 \leq y_j \leq 1$ and $x_{ij} \in \{0, 1\}$.

- Summing (8) over i gives (5); hence every LP solution of (M2) is an LP solution of (M1), and $z^{LP}_{(M2)} \leq z^{LP}_{(M1)}$.
- (M2) has the same variables as (M1) and $mn - n$ additional constraints (mn constraints (8) replace the n constraints (5)).
- In the paper, (M2) is the formulation solved with CPLEX 10.2.

Remark *(implementation)*

The implementation keeps $y_j \in \{0, 1\}$ in the integer version of (M2). By Proposition 3 this does not change the optimal value, and the LP relaxation is the same, since x and y are relaxed to $[0, 1]$ in both cases.

Model M3

For each product j, let $S_{pj}, p = 1, \ldots, k_j$, be the sets of customers with $|S_{pj}| \geq O_j$ and $\sum_{i \in S_{pj}} c_{ij} \leq B_j$. The binary variable z_{pj} equals 1 if product j is offered exactly to S_{pj}.

$$\max \sum_j \sum_p \left[\sum_{i \in S_{pj}} (p_{ij} - c_{ij}) - f_j \right] z_{pj} \tag{9}$$

$$\text{s.t.} \sum_j \sum_p \left[\sum_{i \in S_{pj}} \big(p_{ij} - (1+R)c_{ij}\big) - (1+R) f_j \right] z_{pj} \geq 0 \tag{10}$$ hurdle rate

$$\sum_j \sum_{p:\, i \in S_{pj}} z_{pj} \leq M_i, \qquad i = 1, \ldots, m \tag{11}$$ offers per customer

$$\sum_{p=1}^{k_j} z_{pj} \leq 1, \qquad j = 1, \ldots, n \tag{12}$$ one set per product

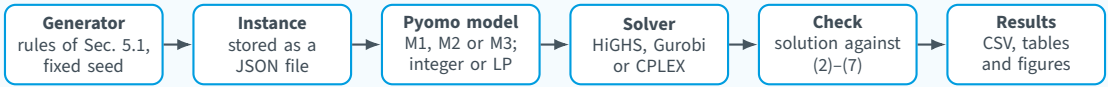
$$z_{pj} \in \{0, 1\}. \tag{13}$$

- Budget (3) and MQC (6) are built into the definition of the columns. (M3) results from a Dantzig–Wolfe decomposition, and $z^\star \leq z^{LP}_{(M3)} \leq z^{LP}_{(M2)}$ [1].
- The number of columns grows as $O(n\, 2^m)$. The paper solves (M3) by column generation within branch-and-price.

Remark *(implementation)*

The implementation enumerates all admissible columns explicitly and is therefore restricted to $m \leq 12$. Branch-and-price is not implemented. The solution of (M3) is mapped back to (x, y) and checked against (2)–(7).

Implementation: Overview



- **Pyomo as a common modelling layer.** Each formulation is written once, in algebraic form close to the paper. The same model is sent to HiGHS, Gurobi or CPLEX by changing only the solver name; Pyomo translates the model and returns the solution, the status and the bound in a uniform way. Only the time limit is passed to the solvers.
- **One model builder for M1 and M2.** The two formulations differ only in the linking constraints, (5) or (8). The LP relaxation is obtained from the same builder by changing the domain of x and y from $\{0, 1\}$ to $[0, 1]$.
- **M3 in two steps.** The admissible columns are enumerated first (budget and MQC checked for each subset of customers), and the set-covering model is then built from them. Its solution is mapped back to (x, y).
- **Safeguards.** Every integer solution, from any model and solver, is checked against (2)–(7) by a function that does not use Pyomo. Unit tests cover the 3-customer example, a brute-force enumeration of all solutions and the reproducibility of the generator.
- **Reproducibility.** Instances are generated from fixed seeds and saved before solving; each run writes one CSV row per instance, model and solver.

Design: Experiments

Experiment 1 *(small instances, M1, M2 and M3)*

- **50 instances**, $m = 4, \ldots, 12$ and $n = 2, \ldots, 5$; R, M_i variant and budget variant cycled deterministically; seeds 4000–4052.
- 53 candidates screened with (M1) and HiGHS; 3 with optimum 0 discarded, because the relative gap is undefined.
- Integer model and LP relaxation of M1, M2, M3; time limit 300 s.
- $50 \times 3 \times 3 = 450$ integer runs and 450 LP runs.

Experiment 2 *(groups S1–S3 of the paper, M1 and M2)*

- **162 instances**: $m = 100$, 200, 300 (S1, S2, S3), each with $n \in \{5, 10, 15\} \times 3$ values of $R \times 2$ M_i variants $\times$ 3 budgets; seed 1.
- (M3) is excluded: explicit enumeration is limited to $m \leq 12$.
- Integer model and LP relaxation of M1 and M2; time limit 3600 s.
- $162 \times 2 \times 3 = 972$ integer runs and 972 LP runs.

Computational environment *(identical for both experiments)*

Machine	Intel Core i7-8565U @ 1.80 GHz (4 cores, 8 threads), 15.8 GB RAM, Windows 10 Home (10.0.19045)
Modelling	Python 3.12.10, Pyomo 6.10.1
Solvers	HiGHS 1.15.1 (`highspy`, in-memory `appsi` interface); Gurobi 13.0.3 (`gurobipy`, LP-file interface); CPLEX 22.2.0.0 (Interactive Optimizer, LP-file interface). Only the time limit is set; all other parameters keep their default values.
Time	wall-clock time of the Pyomo `solve` call: includes the interface overhead, excludes model construction and the enumeration of the M3 columns.

Correctness: every integer solution is checked against (2)–(7) by an independent function. The 10 unit tests pass, including a 3-customer example with 8 admissible columns and a unique optimum 11, confirmed by enumerating all $2^6 = 64$ offer matrices.

Experiment 1: LP bounds

- All 450 integer runs and 450 LP runs end with proven optimality. For every instance, the integer optimum is the same for the three formulations and the three solvers (largest difference 6×10^{-13}).
- The LP gap $100\,(z^{LP} - z^\star)/z^\star$ is identical across solvers (largest difference 9×10^{-14} percentage points); it depends on the formulation only. The values below use the Gurobi runs.
- The ordering $z^\star \le z^{LP}_{(M3)} \le z^{LP}_{(M2)} \le z^{LP}_{(M1)}$ holds in all 50 instances; the inequality between M2 and M1 is strict in 50/50, and between M3 and M2 in 16/50.

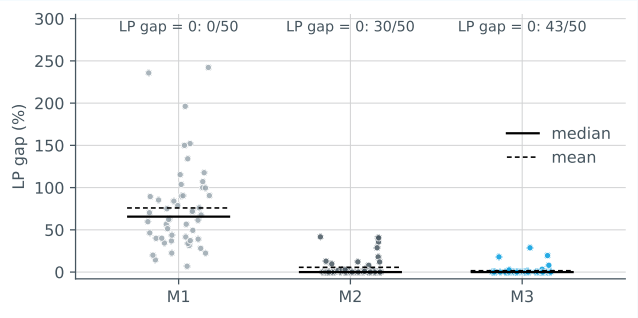


Figure 1. LP gap of M1, M2 and M3 on the 50 instances of Experiment 1; one point per instance, identical for the three solvers.

	LP gap (%)			LP gap
Model	mean	median	max	$= 0$
M1	75.90	65.63	242.22	0/50
M2	5.63	0.00	41.67	30/50
M3	1.63	0.00	28.75	43/50

In 43 of 50 instances the LP relaxation of (M3) already gives the integer optimum; (M2) does so in 30.

Experiment 1: Size of M3

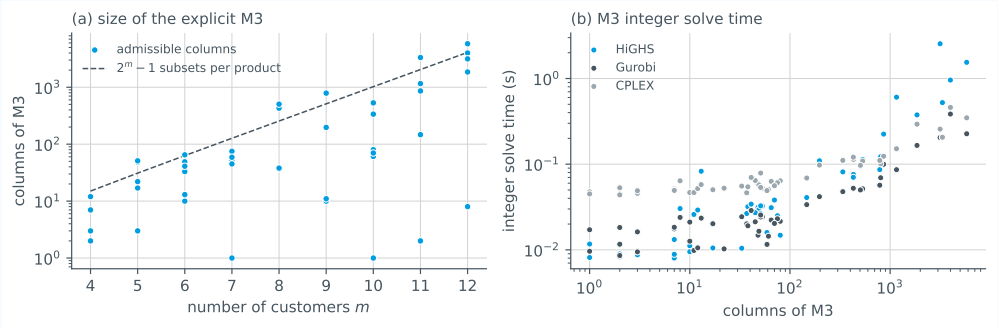


Figure 2. (a) Admissible columns of M3 per instance, as a function of m. (b) Integer solve time of M3 against its number of columns, per solver. Logarithmic scales.

- (a) The number of admissible columns ranges from 1 to 5 809 (median 50). It is bounded by $n(2^m - 1)$, but the budget and the MQC remove a large and variable share of the subsets, so instances with the same m differ by several orders of magnitude.
- (b) The integer solve time of (M3) increases with the number of columns: Spearman correlation 0.88 (HiGHS), 0.81 (Gurobi), 0.88 (CPLEX). All times stay below 2.6 s.

Experiment 1: Solve times

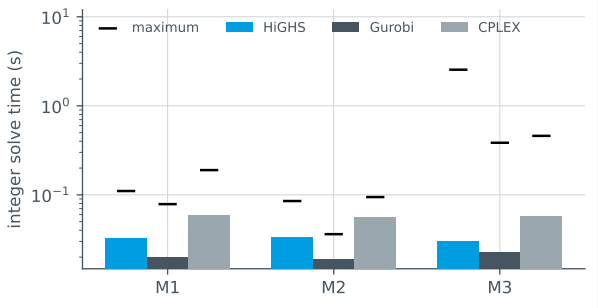


Figure 3. Integer solve time by solver and formulation in Experiment 1: bars show the median over 50 instances, marks the maximum. Logarithmic scale.

		integer time (s)		
Solver	Model	median	mean	max
HiGHS	M1	0.033	0.036	0.110
	M2	0.033	0.033	0.085
	M3	0.030	0.166	2.546
Gurobi	M1	0.020	0.022	0.079
	M2	0.019	0.020	0.036
	M3	0.023	0.049	0.385
CPLEX	M1	0.060	0.064	0.190
	M2	0.056	0.058	0.094
	M3	0.058	0.092	0.460

- The medians are similar across formulations. On the 6 instances with at least 1 000 columns, the median time of (M3) is 0.78 s (HiGHS), 0.21 s (Gurobi) and 0.28 s (CPLEX), against at most 0.07 s for (M1) and (M2).

Remark *(interpretation)*

At this scale, the differences between solvers are of the order of hundredths of a second. They are likely to reflect the interface (in-memory for HiGHS, LP file for Gurobi and CPLEX) more than the algorithms, so they do not rank the solvers.

Experiment 1: Conclusions

1. **The three formulations are correctly implemented.** They reach the same proven optimum on all 50 instances with the three solvers, and every solution satisfies (2)–(7).
2. **The theoretical ordering of the LP bounds is confirmed empirically**: $z^\star \leq z^{LP}_{(M3)} \leq z^{LP}_{(M2)} \leq z^{LP}_{(M1)}$ in all instances.
3. **Disaggregation is the main gain.** The mean LP gap falls from 75.90% (M1) to 5.63% (M2); the set-covering bound reduces it further to 1.63%, and closes it completely in 43 of 50 instances.
4. **The strength of (M3) has a size cost.** The explicit model can have thousands of columns already for $m \leq 12$, and its solve time grows with the number of columns. For realistic m it requires column generation, as in the paper.
5. **The LP bound depends on the formulation, not on the solver**; the solve times of these small instances do not discriminate between solvers.

Experiment 2: Optimality

- All 972 integer runs (162 instances $\times$ 2 models $\times$ 3 solvers) and all 972 LP runs end with status optimal. No run reaches the time limit of 3600 s; the longest takes 1 145 s.
- In 969 of the 972 integer runs the final MIP gap is zero. Three runs on S3 with $n = 15$, $R = 15\%$ and the ℓ variant stop at the default relative tolerance of the solver (0.01%), one unit away from the bound:
  - HiGHS on (M1), two instances: the incumbent is the optimum, but the bound is not closed;
  - Gurobi on (M2), one instance: the incumbent is 10 898, while the other five runs prove 10 899.
- For every other instance, the six runs return the same objective value. The optimum ranges from 488 to 3 446 in S1, 1 477 to 6 455 in S2 and 1 227 to 11 024 in S3.
- As reference optimum $z^\star$ of each instance we take the best objective over its six runs. In every run, $z^\star \leq z^{LP}_{(M2)} \leq z^{LP}_{(M1)}$, and the LP values agree across solvers (largest difference 1.2×10^{-11}).

Remark *(interpretation)*

The three runs above are correct under the default settings of the solvers. They show that status optimal means optimal within a tolerance, which becomes visible when the objective exceeds 10^4.

Experiment 2: LP bounds

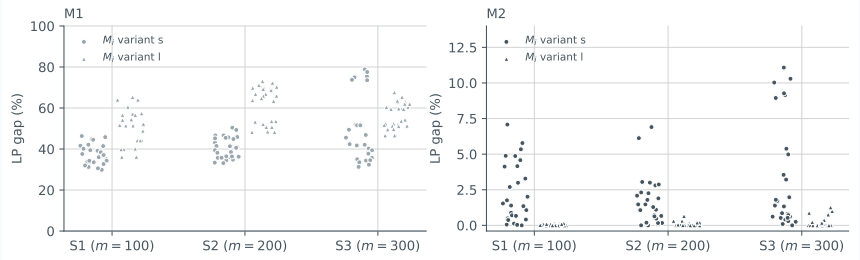


Figure 4. LP gap $100\,(z^{LP} - z^{\star})/z^{\star}$ of M1 (left) and M2 (right) on the 162 instances, by group and M_i variant. The vertical scales differ.

Group	M1: LP gap (%) mean	min	max	M2: LP gap (%) mean	max	mean, s	mean, ℓ
S1 ($m = 100$)	44.36	29.89	65.30	1.18	7.08	2.32	0.04
S2 ($m = 200$)	51.61	33.14	73.09	0.87	6.90	1.66	0.08
S3 ($m = 300$)	52.24	31.29	78.82	1.76	11.08	3.33	0.18

- (M2) closes the LP gap completely in 17, 15 and 17 of the 54 instances of S1, S2 and S3; (M1) never does.
- The gap of (M2) is concentrated in the s variant of M_i; no monotone trend with m is visible across the three groups.

Experiment 2: Solve times

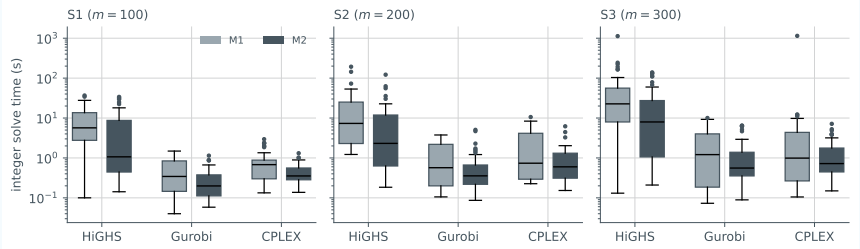


Figure 5. Integer solve time of M1 and M2 per solver, for each group (54 instances per box). Logarithmic scale.

Solver	Model	S1 ($m = 100$) median	S1 ($m = 100$) max	S2 ($m = 200$) median	S2 ($m = 200$) max	S3 ($m = 300$) median	S3 ($m = 300$) max
HiGHS	M1	5.68	36.71	7.33	193.25	22.65	1 128.02
	M2	1.07	33.85	2.32	122.02	7.97	139.14
Gurobi	M1	0.34	1.49	0.57	3.76	1.21	10.02
	M2	0.20	1.14	0.36	5.06	0.56	6.50
CPLEX	M1	0.68	2.95	0.74	10.59	0.99	1 145.38
	M2	0.35	1.31	0.60	6.21	0.72	7.15

- Gurobi is the fastest solver in most instances (M2: 46, 50 and 43 of 54 in S1, S2, S3); CPLEX in the others.
- Geometric mean time relative to Gurobi on (M2): HiGHS 8.67, 6.84, 8.66; CPLEX 1.92, 1.69, 1.29 in S1, S2, S3.

Experiment 2: Effect of the number of customers

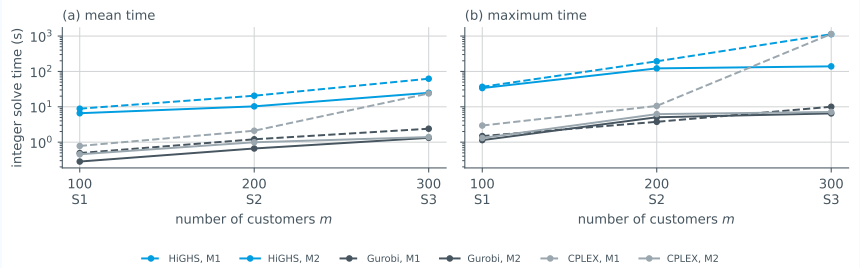


Figure 6. Mean (a) and maximum (b) integer solve time over the 54 instances of each group, per solver and model. Logarithmic scale.

- From S1 to S3, the mean time of (M2) grows from 0.28 to 1.32 s with Gurobi, from 0.45 to 1.39 s with CPLEX and from 6.58 to 24.82 s with HiGHS.
- The maximum grows much faster for (M1): the two longest runs of the experiment are (M1) on S3, with CPLEX (1 145 s) and HiGHS (1 128 s). With (M2), no run exceeds 139 s (HiGHS) or 7.2 s (Gurobi, CPLEX).
- The s variant of M_i remains the harder case: mean (M2) time with HiGHS 12.30, 19.22, 42.71 s for s against 0.87, 1.36, 6.93 s for ℓ.

Remark *(interpretation)*

With three values of m and one seed, these curves describe this sample only and do not support an extrapolation to larger groups.

Experiment 2: M1 versus M2

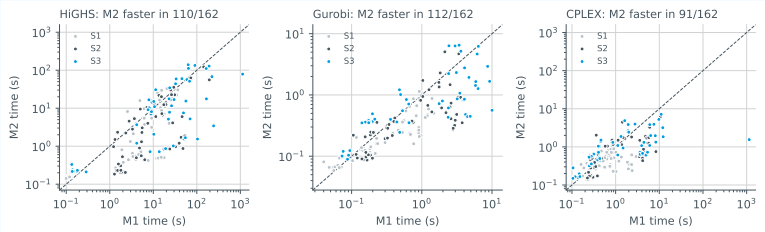


Figure 7. Integer solve time of M2 against M1 for each instance, per solver and group (logarithmic scales). Points below the diagonal are instances solved faster with M2.

	instances where M2 is faster			geometric mean of time(M2)/time(M1)		
Group	HiGHS	Gurobi	CPLEX	HiGHS	Gurobi	CPLEX
S1	35/54	44/54	36/54	0.46	0.66	0.69
S2	43/54	41/54	34/54	0.31	0.59	0.65
S3	32/54	27/54	21/54	0.44	0.71	0.75

Remark *(interpretation)*

(M2) is faster on average in every group and with every solver, but not on every instance; in S3 it is faster in only 27 (Gurobi) and 21 (CPLEX) of 54 instances, although its worst case is much shorter. The solvers presumably strengthen (M1) by presolve and cuts; the logs needed to confirm this were not recorded.

Experiment 2: Conclusions

1. **Groups S1 to S3 are solvable to optimality on a laptop.** All 162 instances are solved with both formulations and all three solvers, within the default tolerance; the longest run takes 1 145 s.
2. **(M2) dominates (M1) in bound quality** in every group: mean LP gap 0.87–1.76% against 44.36–52.24%, consistent with Experiment 1 and Proposition 3.
3. **(M2) is faster on average and more robust.** Its geometric mean time ratio to (M1) is 0.31–0.75, and its worst case stays at 139 s, against about 1 100 s for (M1) on S3.
4. **The solver matters more than the formulation in the typical case**: HiGHS is about 7 to 9 times slower than Gurobi on (M2); CPLEX is 1.3 to 1.9 times slower, with the difference shrinking as m grows.
5. **The s variant of M_i is the harder case** in all groups: larger LP gaps of (M2) and longer solve times.

Summary: Conclusions and limitations

Conclusions

- The implementation reproduces the three exact formulations of the paper, validated by unit tests, an independent feasibility check and the agreement of 3 formulations $\times$ 3 solvers on every optimum.
- The empirical ordering of the LP bounds matches the theory: (M3) is the strongest and (M1) by far the weakest. The disaggregated linking constraints (8) account for most of the improvement.
- Up to $m = 300$, the compact model (M2) with a commercial MIP solver suffices for optimality within the default tolerance: at most 7.2 s per instance with Gurobi or CPLEX.

Limitations

- (M3) is solved only by explicit enumeration ($m \leq 12$); column generation and branch-and-price, which make (M3) usable for $m = 100$, are not implemented.
- Only groups S1 to S3 were run; the larger groups M1, M2 and L ($m = 1\,000$ to $10\,000$) remain untested.
- The generator has two documented deviations and shares data across instances and groups; one seed only per configuration.
- Times are end-to-end Pyomo times under different interfaces, with default solver settings and no solver logs or node counts.

Thank you

Questions?

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Homework 3

Reference

[1] F. Talla Nobibon, R. Leus, F.C.R. Spieksma. Optimization models for targeted offers in direct marketing: Exact and heuristic algorithms. *European Journal of Operational Research*, 210(3):670–683, 2011.